The documentation and process conversion measures necessary to comply with this amendment shall be completed by 29 Jul 1994.

INCH-POUND

MIL-S-19500/304C
AMENDMENT 3
31 March 1994
SUPERSEDING
AMENDMENT 2
17 November 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
FAST-RECOVERY
TYPES 1N3885, 1N3886, 1N3888, 1N3890, 1N3891,
1N3893, 1N3890R, 1N3891R, 1N3893R AND A-VERSIONS
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-S-19500/304C, dated 6 June 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title, delete and substitute as printed below:

"SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
FAST-RECOVERY
TYPES 1N3885, 1N3886, 1N3888, 1N3890, 1N3891,
1N3893, 1N3890R, 1N3891R, 1N3893R AND A-VERSIONS
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC"

* 1.1, end of paragraph, add: " Two levels of product assurance are provided for die."

* 1.2, delete "and 2 (DO-4)" and substitute ", 2 (DO-4), and 2A (die)."

1.3, maximum ratings, following title, add: "$T_C$ = 25°C unless otherwise specified."

* Beneficial comments note, delete address and substitute the following: "US Army Research Laboratory Electronics and Power Sources Directorate, ATTN: AMSRL-EP-RD, Ft. Monmouth, NJ 07703-5601".

PAGE 2

1.4, primary electrical characteristics, following title, add: "$T_C$ = 25°C unless otherwise specified."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

| Replacement page | Page replaced |
|---|---|
| 19 | 19 |
| 20 | 20 |

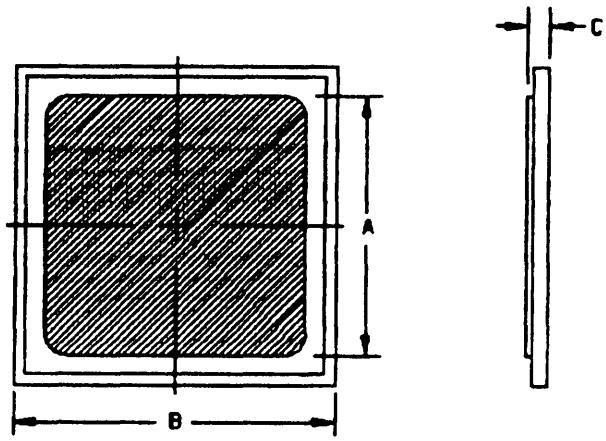

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PAGE 6

* Add the following new figure 2A:
"

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .114 | .117 | 2.89 | 2.97 |
| B | .130 | .133 | 3.30 | 3.37 |
| C | .009 | .010 | .228 | .254 |




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are: Metallization {front (anode), back (cathode)} consists of: Ag thickness = 3,000 Å minimum, NI thickness = 1,500 Å minimum, Cr thickness = 800 Å minimum.
4. Requirements in accordance with MIL-S-15000 (appendix H) are performed in a DO-4 package.

FIGURE 2A. JANHCA and JANKCA die dimensions."

PAGE 7

* Add the following new paragraph:

"4.2.1 Qualification for JANHC and JANKC. Qualification for JANHC and JANKC shall be in accordance with MIL-S-19500."

* 3.3, delete " herein." and substitute "(DO-4), and 2A (die) herein."

4.3.1: Delete and substitute as follows:

"4.3.1 Thermal response. The thermal response measurements shall be performed in accordance with method 3101 of MIL-STD-750. The thermal response conditions and maximum thermal response limit shall be derived by each vendor. The chosen thermal response measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen thermal response values shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. Heating current ($I_H$) ≥ rated $I_O$; $t_H$ = 150 to 400 ms; $t_{MD}$ = 50 to 300 μs; 50 mA ≤ $I_M$ ≤ 250 mA."



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* 4.3.2, delete and substitute:

"4.3.2 Surge current. See MIL-STD-750, method 4066. $I_O$ = 0; $V_{RM}(V)$ = 0; $T_A$ = 25°C; 6 surges. $t_p$ = 8.3 ms, one surge per minute maximum.

| Surge current value ($I_{FSM}$) | | |
|---|---|---|
| | A-version | Non-A version |
| 1N3885 | 290 | 190 |
| 1N3886 | 290 | 190 |
| 1N3888 | 290 | 190 |
| 1N3890 | 310 | 210 |
| 1N3891 | 310 | 210 |
| 1N3893 | 310 | 210 |

"

PAGE 9

* 4.3.4, add the following new paragraph:

"4.3.4 Screening JANHC and JANKC. Screening shall be in accordance with MIL-S-19500 (appendix H). As a minimum, die shall be 100 percent probed to insure compliance with Table I, subgroup 2 herein."

PAGE 10

4.5.4, first line: Delete "30 s ≤ $t_{heating}$ ≤ 60 s" and substitute "$t_{heating}$ ≤ 5 minutes".

PAGE 11

TABLE I, subgroup 2: Delete the following:

| | | | | | | |
|---|---|---|---|---|---|---|
| "Forward voltage | 4011 | $I_F = I_{(FSM)}$<br>$t_p$ = 800 µs | | $V_{F1}$ | 2.75 | V dc |

TABLE I, add the following.

| | | | | | | |
|---|---|---|---|---|---|---|
| "Subgroup 6 | | | | | | |
| Forward voltage 2/ | 4011 | $I_F = I_{(FSM)}$<br>$t_p$ = 800 µs | | $V_{F1}$ | 2.75 | V dc |
| Forward voltage 2/ | 4011 | $I_F = I_{(FSM)}$<br>$t_p$ = 8.3 ms | | $V_{F1}$ | 2.55 | V dc |

"2/ $V_{F1}$ shall be performed with either $t_p$ = 800 µs or $t_p$ 8.3 ms".

PAGE 13

* TABLE IIa, subgroup 6, thermal resistance, conditions column: Delete in its entirety and substitute "$R_{\theta JC}$ = rated $R_{\theta JC}$ (see 1.4); $I_M$ = 50 mA ≤ $I_M$ ≤ 250 mA; $t_H$ = 30 seconds minimum; $t_{MD}$ = 50 to 300 µs".



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PAGE 18

* TABLE IV, subgroup 4, thermal resistance, conditions column: Delete in its entirety and substitute "$R_{\theta JC}$ = rated $R_{\theta JC}$ (see 1.4); $I_M$ = 50 mA to 250 mA; $I_H \geq$ rated $I_O$; $t_H$ = 30 seconds minimum; $t_{MD}$ = 50 to 300 $\mu$s".

PAGE 22

* Add the following new paragraph:

"6.5 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA1N3885) will be identified on the QPL.

| JANHC and JANKC ordering information | |
|---|---|
| PIN 1/ | Manufacturer |
| | 59377 |
| | 1/ 2/ |
| 1N3885 | JANHCA1N3885 |
| 1N3886 | JANHCA1N3886 |
| 1N3888 | JANHCA1N3888 |
| 1N3890 | JANHCA1N3890 |
| 1N3891 | JANHCA1N3891 |
| 1N3893 | JANHCA1N3893 |

1/ Also applies to "A" suffix devices.
2/ For JANS level, replace "JANHC" prfix with "JANKC"."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1507)



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TABLE V. Groups A, B, and C electrical end-point measurements.

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Forward voltage | 4011 | $I_F$ = 38 a (pk) (pulse); pulse width = 8.3 ms duty cycle = 2% maximum | $V_{F2}$ | | 1.50 | V (pk) |
| 2. | Reverse current | 4016 | DC method $V_R$ = Rated $V_R$ (dc) (see 1.3) | $I_{R1}$ | | 10 | µA dc |
| 3. | Thermal response | 3101 | See 4.3.1 $R_{\theta JC}$= rated $R_{\theta JC}$ (see 1.4) | $\Delta V_F$ | | | mV dc |
| 4. | Forward voltage | 4011 | $I_F$ = 50 mA dc | $\Delta V_{F3}$ | +50 mV dc maximum change from previous to post intermittent life and thermal shock measurements tests (JANS only) | | |
| 5. | Reverse current | 4016 | DC method $V_R$ = Rated $V_R$ (dc) (see 1.3) | $\Delta I_{R1}$ 1/ | 100% or ±5 µA (whichever is greater), change from initial group A reading | | |
| 6. | Reverse current | 4016 | DC method $V_R$ = Rated $V_R$ (dc) (see 1.3) | $I_{R3}$ | | 20 | µA dc |

1/ Devices which exceed the group A limits from this test shall be rejected.



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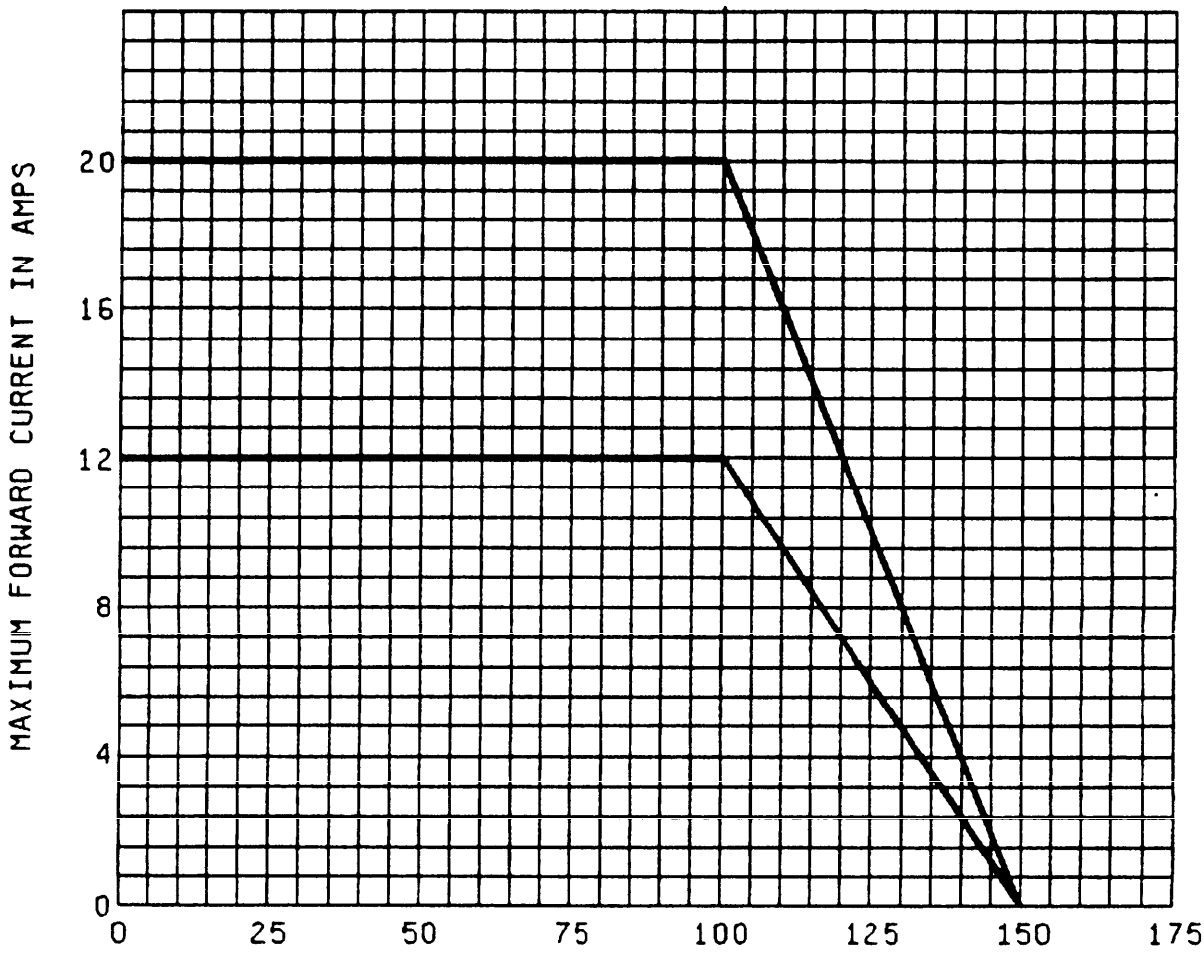



FIGURE 3. Maximum case temperature in °C.



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